Chapter 6: Life Processes - Photosynthesis and Respiration

1. Introduction to Photosynthesis

Photosynthesis is the process by which green plants, algae and some bacteria make their own food using sunlight, water and carbon dioxide. It takes place mainly in the leaves of a plant, inside special structures called chloroplasts. Chloroplasts contain a green pigment called chlorophyll, which absorbs light energy, mostly from the red and blue parts of the spectrum. Green light is reflected, which is why leaves look green to us. The food made by photosynthesis is glucose, a simple sugar, and oxygen is released as a by-product.

The overall reaction is often written as a word equation: carbon dioxide + water + light energy gives glucose + oxygen. In chemical symbols this is 6CO2 + 6H2O + light -> C6H12O6 + 6O2. Photosynthesis is important because nearly all food chains begin with it, and because it replaces the oxygen that living things use up. Dr. Joseph Priestley showed in 1771 that a plant kept in a closed jar could restore air in which a candle had burned out, e.g. a mint plant revived the air after a candle had used it up. Later, Jan Ingenhousz found that this only worked in sunlight.

1.1 Where photosynthesis happens

A typical leaf has an upper epidermis, a palisade layer, a spongy layer and a lower epidermis. The palisade cells are packed with chloroplasts and sit just under the upper surface so they receive the most light (see Fig. 1). Tiny pores called stomata, mostly on the lower surface, let carbon dioxide enter and oxygen leave. Each stoma is guarded by two kidney-shaped guard cells that open and close the pore. Water reaches the leaf from the roots through the xylem, and the glucose that is made leaves through the phloem.

- Chloroplast: the organelle where photosynthesis occurs
- Chlorophyll: the green pigment that absorbs light
- Stomata: pores that control gas exchange
- Xylem: carries water and minerals upwards
- Phloem: carries food (sucrose) to all parts of the plant

2. The Two Stages of Photosynthesis

Photosynthesis happens in two linked stages. The first is the light-dependent stage, which takes place in the thylakoid membranes of the chloroplast. Here chlorophyll absorbs light and the energy is used to split water molecules into hydrogen ions, electrons and oxygen. This is called photolysis of water. The oxygen is released through the stomata. The energy captured is stored temporarily in two molecules, ATP and NADPH, which act as short-term energy carriers for the next stage.

The second stage is the light-independent stage, also known as the Calvin cycle. It takes place in the stroma, the fluid that fills the chloroplast. Carbon dioxide is fixed, that is, attached to a five-carbon sugar called RuBP with the help of the enzyme RuBisCO. Using the ATP and NADPH from the first stage, the resulting molecules are converted into glucose. The Calvin cycle does not need light directly, but it stops soon after dark because the supply of ATP and NADPH runs out. Approx. 3 molecules of CO2 must enter the cycle to make one three-carbon sugar, and six turns are needed to build one glucose molecule.

1. Light is absorbed by chlorophyll in the thylakoid membranes.
2. Water is split (photolysis), releasing oxygen, electrons and hydrogen ions.
3. ATP and NADPH are produced and carried to the stroma.
4. Carbon dioxide is fixed by RuBisCO in the Calvin cycle.
5. Glucose is built and either used, stored as starch, or transported as sucrose.

2.1 What happens to the glucose

The glucose made in the leaf has several possible fates. Some is used immediately in respiration to release energy. Some is converted to starch and stored in the leaf, in roots, in tubers such as potatoes, and in seeds. Some is changed to sucrose for transport, and some is combined with nitrogen from the soil to make proteins, or used to make cellulose for cell walls. A simple way to test a leaf for starch is the iodine test: starch turns iodine solution from brown to blue-black. Before testing, the leaf is boiled in water and then in alcohol to remove the chlorophyll so the colour change can be seen clearly.

3. Factors Affecting the Rate of Photosynthesis

The rate of photosynthesis depends on several factors, and the one in shortest supply at any moment is called the limiting factor. This idea was first stated by F. F. Blackman in 1905. When light is dim, increasing light intensity increases the rate almost in proportion, so light is limiting. At higher intensities the rate levels off because another factor, such as carbon dioxide concentration or temperature, becomes limiting instead. Plotted on a graph, this gives a curve that rises steeply and then flattens (Fig. 2).

- Light intensity: more light increases the rate until another factor limits it
- Carbon dioxide concentration: normal air contains approx. 0.04% CO2, which is low, so adding CO2 often raises the rate
- Temperature: enzymes work best at about 25 to 35 degrees Celsius; above approx. 45 degrees they denature and the rate falls sharply
- Water supply: severe shortage closes the stomata and stops CO2 entering
- Chlorophyll amount: leaves with less chlorophyll, e.g. in yellowing leaves, photosynthesise more slowly

Farmers and greenhouse growers use this knowledge. By raising CO2 levels, keeping a steady warm temperature and providing artificial light, they can increase yields, i.e. the amount of crop produced per unit area. However, the cost of heating and lighting must be weighed against the extra crop, which is why growers only add the factor that is limiting at that time.

4. Cellular Respiration

Respiration is the process by which cells break down glucose to release energy that can be used for life activities such as movement, growth, active transport and keeping warm. It is not the same as breathing; breathing is the exchange of gases, while respiration is a chemical process that happens in every living cell, all the time, day and night. The energy is not released as heat all at once. Instead, it is captured step by step in a molecule called ATP, which cells can use wherever energy is needed. Each molecule of ATP can be thought of as a small rechargeable battery.

4.1 Aerobic respiration

Aerobic respiration needs oxygen. Glucose is first broken down into two molecules of pyruvate in the cytoplasm, a stage called glycolysis, which does not need oxygen. If oxygen is available, the pyruvate enters the mitochondria, where it is completely oxidised in the Krebs cycle and the electron transport chain. The products are carbon dioxide, water and a large amount of energy, about 30 to 32 molecules of ATP per molecule of glucose. The word equation is glucose + oxygen gives carbon dioxide + water + energy, and the symbol equation is C6H12O6 + 6O2 -> 6CO2 + 6H2O + energy.

4.2 Anaerobic respiration

When oxygen is not available, cells can still release a little energy by anaerobic respiration. In yeast, the pyruvate is converted to ethanol and carbon dioxide, which is called alcoholic fermentation and is used in baking and brewing. In our muscle cells during very hard exercise, pyruvate is converted to lactic acid.

Anaerobic respiration releases only 2 molecules of ATP per glucose, far less than aerobic respiration, which is why it can only support short bursts of activity. The lactic acid that builds up causes muscle fatigue and cramps, and it is broken down later when oxygen becomes available again; the extra oxygen needed is called the oxygen debt.

5. Comparing Photosynthesis and Respiration

Photosynthesis and respiration are often described as opposite processes, and in a way they are, because the products of one are the raw materials of the other. Photosynthesis stores energy in glucose and takes in carbon dioxide, while respiration releases that energy and gives carbon dioxide back. However, they are not simply the reverse of each other: they happen in different organelles and use different enzymes and many different steps.

- Photosynthesis happens only in cells with chloroplasts; respiration happens in all living cells
- Photosynthesis needs light; respiration goes on in light and in darkness
- Photosynthesis stores energy; respiration releases energy
- Photosynthesis takes in CO2 and gives out O2; respiration takes in O2 and gives out CO2 (aerobic)
- Photosynthesis increases the dry mass of a plant; respiration decreases it

During the day, a plant photosynthesises faster than it respires, so overall it takes in carbon dioxide and gives out oxygen. At night there is no light, so photosynthesis stops while respiration continues and the plant gives out carbon dioxide. At one particular light intensity, called the compensation point, the two rates are exactly equal and there is no net exchange of gases. Plants growing in deep shade have a lower compensation point than plants in full sun.

6. Experiments and Practical Skills

Several simple experiments can be done in a school laboratory. To show that light is necessary, a plant is kept in the dark for 48 hours to destarch it, then part of a leaf is covered with black paper and the plant is placed in sunlight for several hours. When tested with iodine, only the uncovered part turns blue-black. To show that carbon dioxide is necessary, one plant is placed in a sealed bell jar with a dish of potassium hydroxide, which absorbs CO2, and a control plant is placed in a jar without it. To measure the rate of photosynthesis, pondweed such as Elodea is placed in water under a lamp, and the bubbles of oxygen released per minute are counted at different distances from the lamp.

When writing up an experiment, always state the aim, the apparatus, the method, the results in a table, a conclusion and possible sources of error. Remember that a control is a set-up that is identical except for the one factor being tested, i.e. the independent variable. Keeping all other variables the same makes the test fair. Examiners frequently award marks for stating what was kept constant, so do not forget this.

7. Revision Questions

1. State the word equation and the balanced symbol equation for photosynthesis. [2 marks]
2. Name the two stages of photosynthesis and state where each takes place. [4 marks]
3. Explain why a plant kept in the dark for two days will not give a blue-black result with iodine. [3 marks]
4. Compare aerobic and anaerobic respiration in terms of oxygen use, products and energy released. [5 marks]
5. Explain what is meant by a limiting factor and describe how light intensity can become limiting. [4 marks]
6. Why does a green plant give out carbon dioxide at night? [2 marks]